

Software Appendix



1. OpenAI API (GPT-based model)

We use large language models via the OpenAI API to generate real-time, Lexile-adaptive explanations, comprehension questions, and personalized feedback based on student input. This enables Lumen to act as an intelligent reading tutor capable of adapting to different grade levels and learning gaps instantly.

- Developed by: OpenAI
- Why used: To power natural-language understanding, adaptive tutoring, and real-time comprehension assessment at scale.

2. Google Cloud (or Firebase backend)

Used for secure storage and syncing of student responses, comprehension checkpoints, and progress tracking across devices for teachers and parents.

- Developed by: Google
- Why used: To enable real-time analytics, scalability, and secure cross-platform data access.

3. Computer Vision (Camera-based text capture via ML models / OCR)

Used to detect and extract text selected by the wand camera, allowing students to highlight passages naturally without manual typing.

- Developed by: Combination of open-source OCR frameworks (e.g., Tesseract / ML libraries)
- Why used: To enable seamless "point-and-understand" reading interaction.